Experiment - 02

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Semester: 5
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Date of Performance: 13/7/26
Subject Code: 24CSP-310

(A)

- **(A)**
  A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

Output

Department	Average Marks
Mathematics	95.00
Computer Science	83.33

Solution:

```
SELECT D.DEPT_NAME AS DEPARTMENT, ROUND(AVG(M.MARKS_SCORED), 2) AS AVERAGE_MARKS
FROM DEPARTMENTS AS D
JOIN STUDENTS AS S ON D.DEPT_ID = S.DEPT_ID
JOIN MARKS AS M ON S.STUDENT_ID = M.STUDENT_ID
GROUP BY D.DEPT_NAME
ORDER BY AVERAGE_MARKS DESC;
```

(B)

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee** (by EmpID) along with their **lowest recorded salary** across both systems.

Objective

1. Combine two tables A and B.
2. Return each EmpID with their **lowest salary,** and the corresponding **Ename.**

Table A

EmpID	Ename	Salary
1	AA	1000
2	BB	300

Table B

EmpID	Ename	Salary
2	BB	400
3	CC	100

Output

EmpID	Ename	Salary
1	AA	1000
2	BB	300
3	CC	100

Solution:

```
SELECT EMPID, MIN(ENAME) AS ENAME, MIN(SALARY) AS SALARY
FROM (
    SELECT * FROM A
    UNION ALL
    SELECT * FROM B
) AS INTERMEDIATE_RES
GROUP BY EMPID;
```

(C)

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.
If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the **Department Name**, **Employee Name**, and **Salary**, arranged in ascending order of department name.

Note: The employee and department information are maintained in separate normalized tables.

Input Table: **Employee**

ID	NAME	SALARY	DEPT_ID
1	JOE	70000	1
2	JIM	90000	1
3	HENRY	80000	2
4	SAM	60000	2
4	MAX	90000	1

Department

ID	DEPT_NAME
1	IT
2	SALES

OUTPUT

DEPT_NAME	NAME	SALARY
IT	MAX	90000
IT	JIM	90000
SALES	HENRY	80000

Part B: FIND THE 2ND HIGHEST EARNING EMPLOYEE from the same schema.

Solution:

(a)

```
SELECT D.DEPT_NAME, E.NAME, E.SALARY
FROM EMPLOYEE AS E
JOIN DEPARTMENT AS D ON E.DEPARTMENT_ID = D.ID
WHERE E.SALARY IN (
    SELECT MAX(SALARY)
    FROM EMPLOYEE AS E1
    WHERE E1.DEPARTMENT_ID = E.DEPARTMENT_ID
)
ORDER BY DEPT_NAME;
```

(b)

```
SELECT D.DEPT_NAME, E.NAME, E.SALARY
FROM EMPLOYEE AS E
JOIN DEPARTMENT AS D ON E.DEPARTMENT_ID = D.ID
WHERE E.SALARY IN (
    SELECT MAX(SALARY)
    FROM EMPLOYEE AS E1
    WHERE E1.DEPARTMENT_ID = E.DEPARTMENT_ID
      AND E1.SALARY NOT IN (
          SELECT MAX(SALARY)
          FROM EMPLOYEE AS E2
          WHERE E2.DEPARTMENT_ID = E1.DEPARTMENT_ID
      )
)
ORDER BY DEPT_NAME;
```